Golden_Kelly_Final_Submission

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Resources Used: Claude (Anthropic) was used in this assignment as noted in the code comments throughout. It was also used to help select genes of personal interest, draft the gene rationale and variable descriptions, improve organization, and debug code errors.

Variables Chosen: My primary gene is *CXCR2*, a receptor on neutrophils that responds to the inflammatory signal IL-8. Inflammation is thought to play a role in preeclampsia, which made *CXCR2* an interesting gene to explore. My continuous covariate is maternal vitamin D (mat_vitD), measured as 25-hydroxyvitamin D in ng/mL at baseline. My two categorical covariates are pregnancy outcome (preg_preeclampsia: normal vs. preeclampsia) and obesity status, which I derived from maternal BMI (mat_BMI) using the WHO cut point of 30 kg/m² or higher.

Additional genes: I added *NR0B2* and *TRIB1* to contrast with *CXCR2*. *NR0B2* is a nuclear receptor, expressed mainly in the liver, that helps regulate bile acid, cholesterol, and glucose metabolism; variants in it have been linked to obesity and early-onset type 2 diabetes. *TRIB1* controls fat production in the liver and is also active in immune cells, so it bridges the other two; large genetic studies have tied it to blood lipid levels and fatty liver disease. Both connect to my broader interest in PFAS, which disrupt nuclear-receptor signaling and are consistently associated with higher cholesterol. Comparing the three lets me see whether the vitamin D and obesity patterns seen for *CXCR2* also appear in metabolic genes.

Heatmap and forest plot: For the heatmap, I selected seven more genes related to inflammation (*LYN*, *PRKCD*, *AQP9*, *FOSL2*) and lipid metabolism (*ACSL4*, *EPHX2*, *PEX12*), choosing ones that varied enough across participants to visualize. For my additional plot, I used a forest plot to test whether obesity explains the association between vitamin D and expression of these ten genes.

Load the data

```
# read_csv() automatically strips the invisible character (BOM) at the start
# of the covariate file, so the first column is named "subjectID" correctly.
covar <- read_csv("/Users/kellygolden/Library/CloudStorage/OneDrive-DartmouthCollege/HSE 103/Final Proj

# gene expression: rows = probe IDs, columns = samples (GEO IDs).
# the first column has no name in the file, so rename it PROBEID.
geneExp <- read_csv("/Users/kellygolden/Library/CloudStorage/OneDrive-DartmouthCollege/HSE 103/Final Pr
  rename(PROBEID = 1)

# linker file: probe ID -> gene name
geneNames <- read_csv("/Users/kellygolden/Library/CloudStorage/OneDrive-DartmouthCollege/HSE 103/Final

dim(covar)
```

```
## [1] 157  10
```

```
dim(geneExp)
```

```
## [1] 500 158
```

Clean the covariates

Each covariate value (except the IDs) has a text label in front of it, like `"age_years: 31...."` or `"pregnancy_condition: Normal"`. I have to remove everything up to and including the `": "` and then convert to the right type of variable.

```
# Look at the raw format first
head(covar$mat_age_yrs, 3)
```

```
## [1] "age_years: 31.06639288" "age_years: 27.79466119" "age_years: 34.31895962"
```

```
# columns 3 to 10 have the "label: value"
for (col in names(covar)[3:10]) {
  covar[[col]] <- sub(".*: ", "", covar[[col]])   # keep only the part after ":"
} # Claude helped with this cleaning code

# convert from text to numbers
numVars <- c("mat_age_yrs", "delivery_gestAge_wks",
             "sampleCollection_gestAge_wks", "mat_BMI", "mat_vitD")
for (col in numVars) {
  covar[[col]] <- as.numeric(covar[[col]])
}

# convert categorical variables to factors with clear labels (Claude helped here)
covar$preg_preeclampsia <- factor(covar$preg_preeclampsia,
                                  levels = c("Normal", "Preeclampsia"))
covar$mat_asthma <- factor(covar$mat_asthma, levels = c(0, 1),
                           labels = c("Healthy", "Asthma"))
covar$batch_Microarray <- factor(covar$batch_Microarray)

# derived categorical variable: obesity status from BMI (WHO: BMI >= 30)
covar$obesity <- factor(ifelse(covar$mat_BMI >= 30, "Obese", "Not obese"),
                        levels = c("Not obese", "Obese"))

# confirm no missing values were created by the conversions (Claude helped here)
colSums(is.na(covar))
```

```
##                     subjectID                         geoID
##                             0                             0
##                  mat_age_yrs             preg_preeclampsia
##                             0                             0
##         delivery_gestAge_wks sampleCollection_gestAge_wks
##                             0                             0
##             batch_Microarray                       mat_BMI
##                             0                             0
##                      mat_vitD                   mat_asthma
##                             0                             0
##                       obesity
##                             0
```

```
summary(covar[, c(numVars, "preg_preeclampsia", "obesity")])
```

```
##   mat_age_yrs    delivery_gestAge_wks sampleCollection_gestAge_wks
##  Min.   :18.43   Min.   :26.43        Min.   :10.00
##  1st Qu.:22.15   1st Qu.:38.14        1st Qu.:12.14
##  Median :26.57   Median :39.29        Median :13.43
##  Mean   :26.38   Mean   :38.82        Mean   :14.13
```

```
##  3rd Qu.:30.14   3rd Qu.:40.00        3rd Qu.:16.43
##  Max.   :38.76   Max.   :42.00        Max.   :18.86
##     mat_BMI         mat_vitD        preg_preeclampsia      obesity
##  Min.   :17.95   Min.   : 5.11   Normal      :110     Not obese:100
##  1st Qu.:22.90   1st Qu.:13.40   Preeclampsia: 47     Obese    : 57
##  Median :27.33   Median :20.40
##  Mean   :28.94   Mean   :22.98
##  3rd Qu.:33.27   3rd Qu.:29.50
##  Max.   :61.95   Max.   :80.80
```

Match the probes to the gene names

```
# Claude helped me choose which functions to apply for this section given
# my R language is still developing

# confirm every probe in the expression data has a gene name in the linker file
all(geneExp$PROBEID %in% geneNames$PROBEID)   # expect TRUE
```

```
## [1] TRUE
```

```
geneExp <- geneExp %>%
  left_join(geneNames, by = "PROBEID") %>%   # add a GENE column
  relocate(GENE, .after = PROBEID)           # move it next to PROBEID

# count duplicated gene names (each gene should be one probe)
sum(duplicated(geneExp$GENE))   # expect 0
```

```
## [1] 0
```

Merge

The expression data has samples as columns, while the covariates have samples as rows. I reshaped the expression data to long format (one row per gene per sample) and joined *CXCR2* to the covariates to verify the merge.

```
gene <- "CXCR2"

geneLong <- geneExp %>%
  select(-PROBEID) %>%
  pivot_longer(cols = -GENE, names_to = "geoID", values_to = "expression")

dat <- geneLong %>%
  filter(GENE == gene) %>%
  inner_join(covar, by = "geoID")
```

Verify

```
# Claude helped with the code below:
# confirm the same samples are in both files
setequal(covar$geoID, unique(geneLong$geoID))   # expect TRUE
```

```
## [1] TRUE
```

```
# confirm no one was lost or duplicated in the join
nrow(dat)                        # expect 157
```

```
## [1] 157
```

```
sum(duplicated(dat$geoID))       # expect 0
```

```
## [1] 0
```

```
# the joined value should match the raw value
check_id <- dat$geoID[1]
raw_value <- geneExp %>% filter(GENE == gene) %>% pull(all_of(check_id))
dat$expression[1] == raw_value   # expect TRUE
```

```
## [1] TRUE
```

Plot theme

```
theme_pub <- theme_classic(base_size = 12) +
  theme(axis.title   = element_text(face = "bold"),
        axis.text    = element_text(color = "#280003"),
        legend.title = element_text(face = "bold"),
        legend.position = "top")

obesityColors <- c("Not obese" = "#4E8098", "Obese" = "#86BA90")
```

Combine all genes with covariates

```
# join all genes to the covariates so the function can pull out any gene
datAll <- geneLong %>%
  inner_join(covar, by = "geoID")

nrow(datAll)   # expect 78500
```

```
## [1] 78500
```

Plotting function

```
plotGenes <- function(dataAll, genes) {
  for (g in genes) {
    geneData <- dataAll %>% filter(GENE == g)

    # histogram
    histPlot <- ggplot(geneData, aes(x = expression)) +
      geom_histogram(bins = 25, fill = "#713E5A", color = "white") +
      labs(x = bquote(bolditalic(.(g)) ~
                      bold("expression (log2 normalized intensity)")),
           y = "Number of participants") +
      scale_y_continuous(expand = expansion(mult = c(0, 0.05))) +  # bars sit on the axis
      theme_pub
    print(histPlot)

    # scatterplot
    scatterPlot <- ggplot(geneData, aes(x = mat_vitD, y = expression)) +
      geom_point(color = "#713E5A", alpha = 0.7, size = 2) +
      geom_smooth(method = "lm", se = TRUE, color = "#280003", linewidth = 0.8) +
      labs(x = "Maternal 25-hydroxyvitamin D (ng/mL)",
```

```
          y = bquote(bolditalic(.(g)) ~ bold("expression (log2)"))) +
      theme_pub
    print(scatterPlot)

    # boxplot
    boxPlot <- ggplot(geneData, aes(x = preg_preeclampsia, y = expression, fill = obesity)) +
      geom_boxplot(outlier.shape = NA, width = 0.6,
                   position = position_dodge(width = 0.7)) +
      # show the participants on top of the boxes (Claude assisted)
      geom_point(position = position_jitterdodge(jitter.width = 0.15,
                                                 dodge.width = 0.7,
                                                 seed = 123),  # added a seed
                 size = 1.2, alpha = 0.5, show.legend = FALSE) +
      scale_fill_manual(values = obesityColors, name = "Obesity status") +
      labs(x = "Pregnancy outcome", y = bquote(bolditalic(.(g)) ~ bold("expression (log2)"))) +
      theme_pub
    print(boxPlot)

  }
}
```

Figure caption

```
# Claude helped figure this code out
# genes to plot
myGenes <- c("CXCR2", "NROB2", "TRIB1")

# build one caption per plot, in the order the function prints them
plotCaps <- c()   # start with an empty list of captions
for (g in myGenes) {
  plotCaps <- c(plotCaps,
    paste0("\\textbf{Distribution of \\textit{", g, "} expression.} ",
         "Expression values from 157 pregnant participants are log2-transformed ",
         "normalized intensities from the Affymetrix Human Gene 1.0 ST Array, ",
         "measured in blood collected at 10--19 weeks' gestation."),
    paste0("\\textbf{Maternal vitamin D vs. \\textit{", g, "} expression.} Each ",
          "point is one participant (n = 157). The black line is a linear ",
          "regression fit, and the gray band is its 95\\% confidence interval."),
    paste0("\\textbf{\\textit{", g, "} expression by pregnancy outcome and obesity status.} ",
       "Obesity was defined as BMI $\\geq$ 30 kg/m². Boxes show the median and ",
           "interquartile range, whiskers extend to 1.5 × IQR, and points are ",
           "individual participants. Group sizes: Normal/not obese n = 79, ",
           "Normal/obese n = 31, Preeclampsia/not obese n = 21, Preeclampsia/obese ",
           "n = 26.")
  )
}

length(plotCaps)   # expect 9
```

```
## [1] 9
```

Box, Histogram, and Scatterplots

```
# run the function
plotGenes(datAll, myGenes)
```

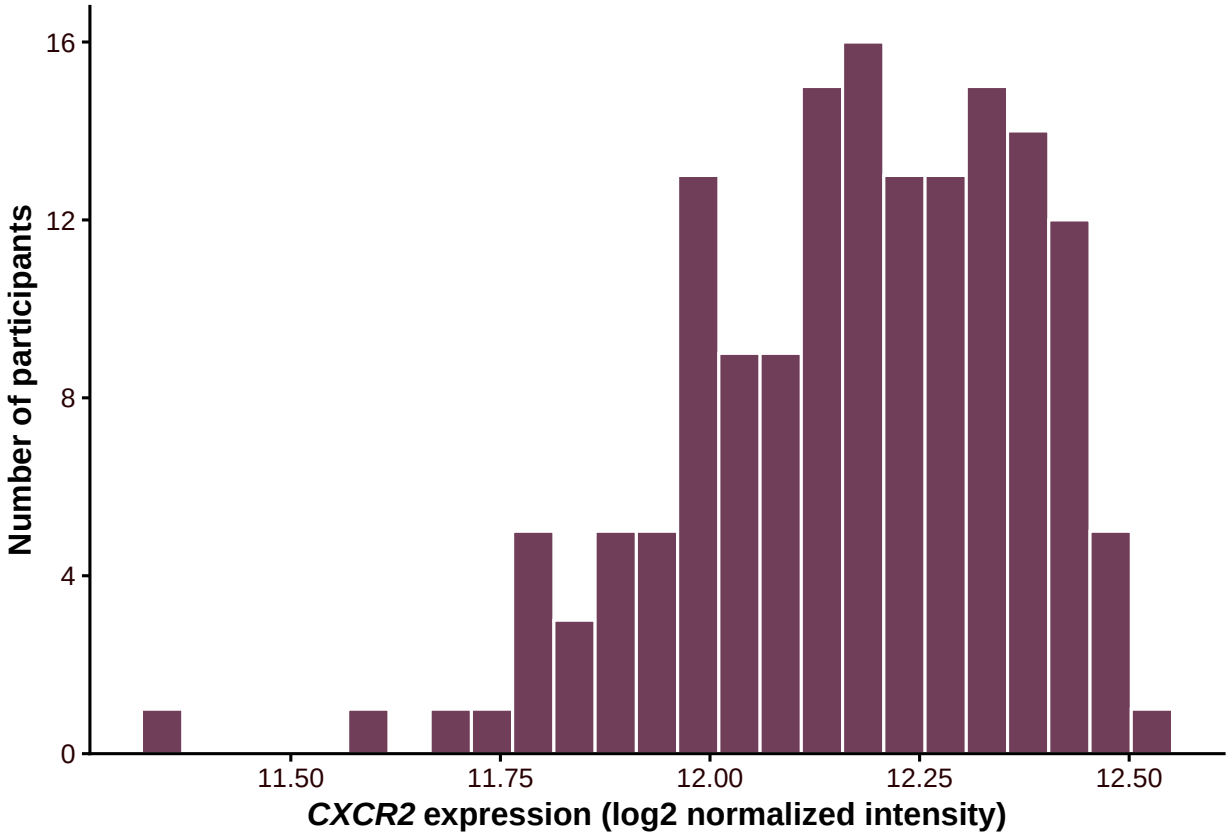


Figure 1: **Distribution of *CXCR2* expression.** Expression values from 157 pregnant participants are log2-transformed normalized intensities from the Affymetrix Human Gene 1.0 ST Array, measured in blood collected at 10–19 weeks' gestation.

Table 1 prep (Summary Statistics)

```
# Claude helped figure this code out

# summary statistics for Table 1: maternal age, gestational age at delivery,
# maternal vitamin D (mean and SD), and obesity (n and %),
# stratified by pregnancy outcome
summaryTable1 <- covar %>%
  group_by(preg_preeclampsia) %>%
  summarise(n = n(),
            age_mean     = mean(mat_age_yrs),
            age_sd       = sd(mat_age_yrs),
            gestAge_mean = mean(delivery_gestAge_wks),
            gestAge_sd   = sd(delivery_gestAge_wks),
            vitD_mean = mean(mat_vitD),
            vitD_sd   = sd(mat_vitD),
            obese_n   = sum(obesity == "Obese"),
```

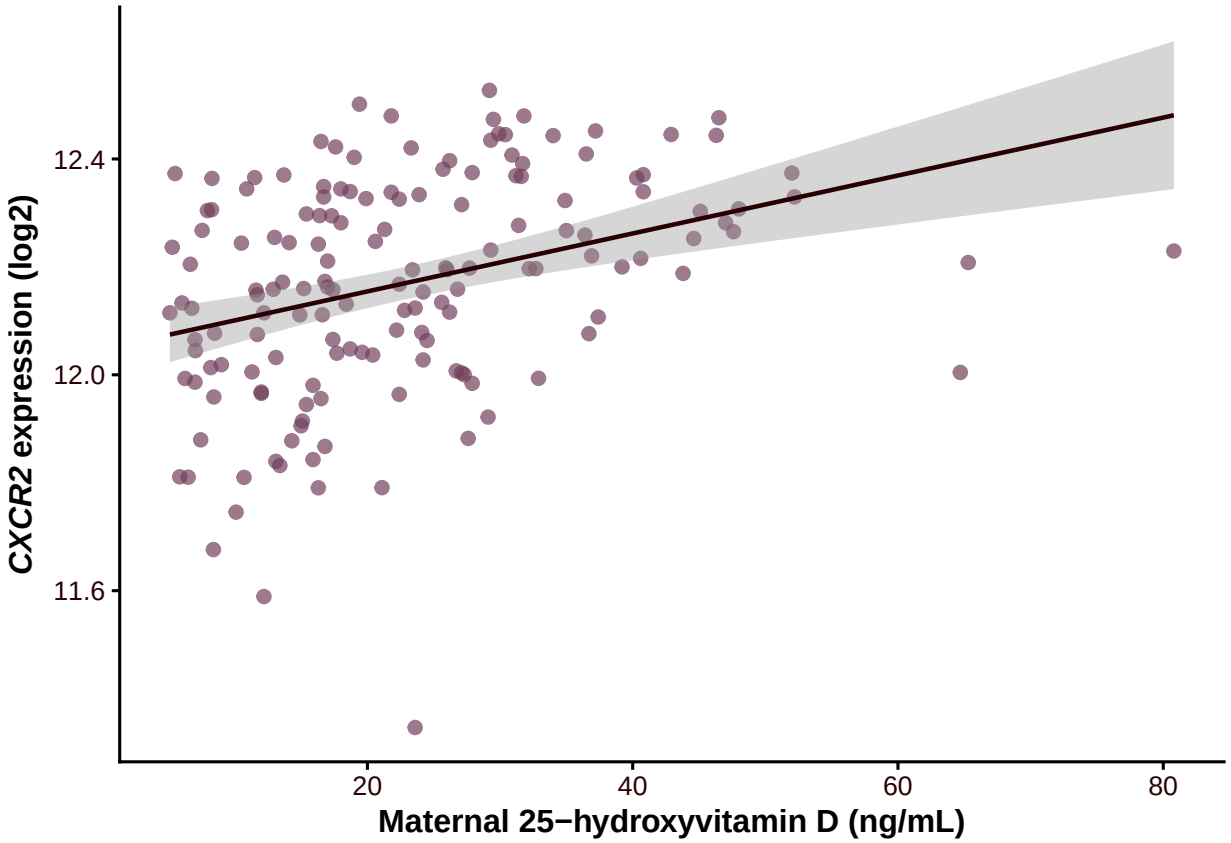


Figure 2: **Maternal vitamin D vs. *CXCR2* expression.** Each point is one participant (n = 157). The black line is a linear regression fit, and the gray band is its 95% confidence interval.

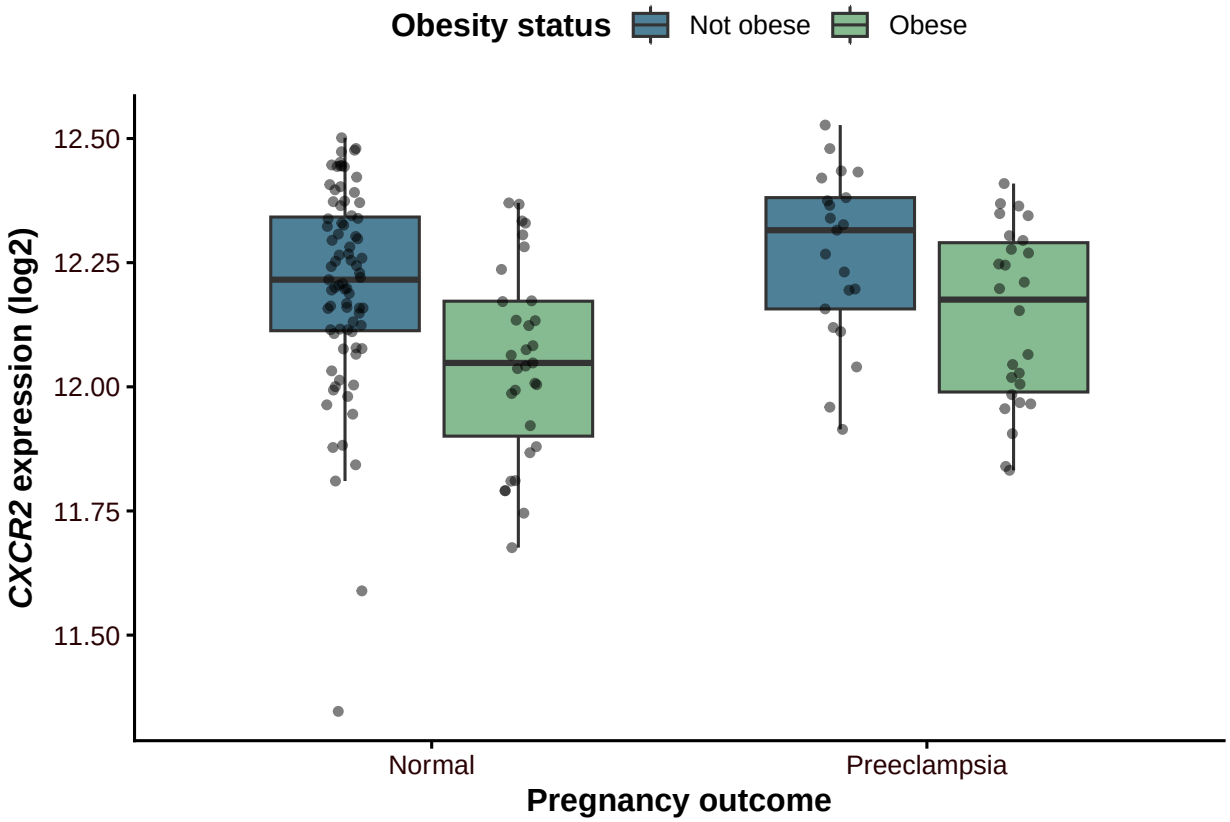


Figure 3: ***CXCR2* expression by pregnancy outcome and obesity status.** Obesity was defined as BMI $\geq$ 30 kg/m^2. Boxes show the median and interquartile range, whiskers extend to 1.5 $\times$ IQR, and points are individual participants. Group sizes: Normal/not obese n = 79, Normal/obese n = 31, Preeclampsia/not obese n = 21, Preeclampsia/obese n = 26.

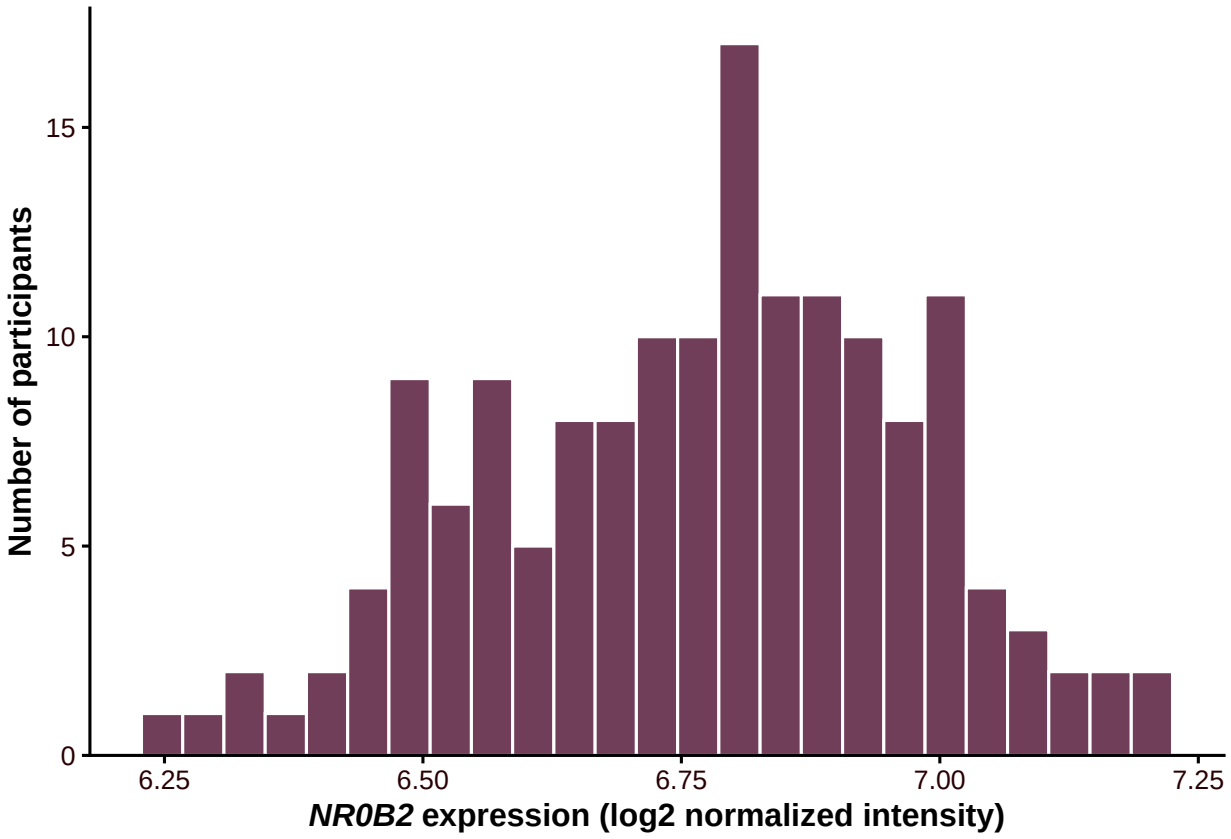


Figure 4: **Distribution of *NR0B2* expression.** Expression values from 157 pregnant participants are log2-transformed normalized intensities from the Affymetrix Human Gene 1.0 ST Array, measured in blood collected at 10–19 weeks' gestation.

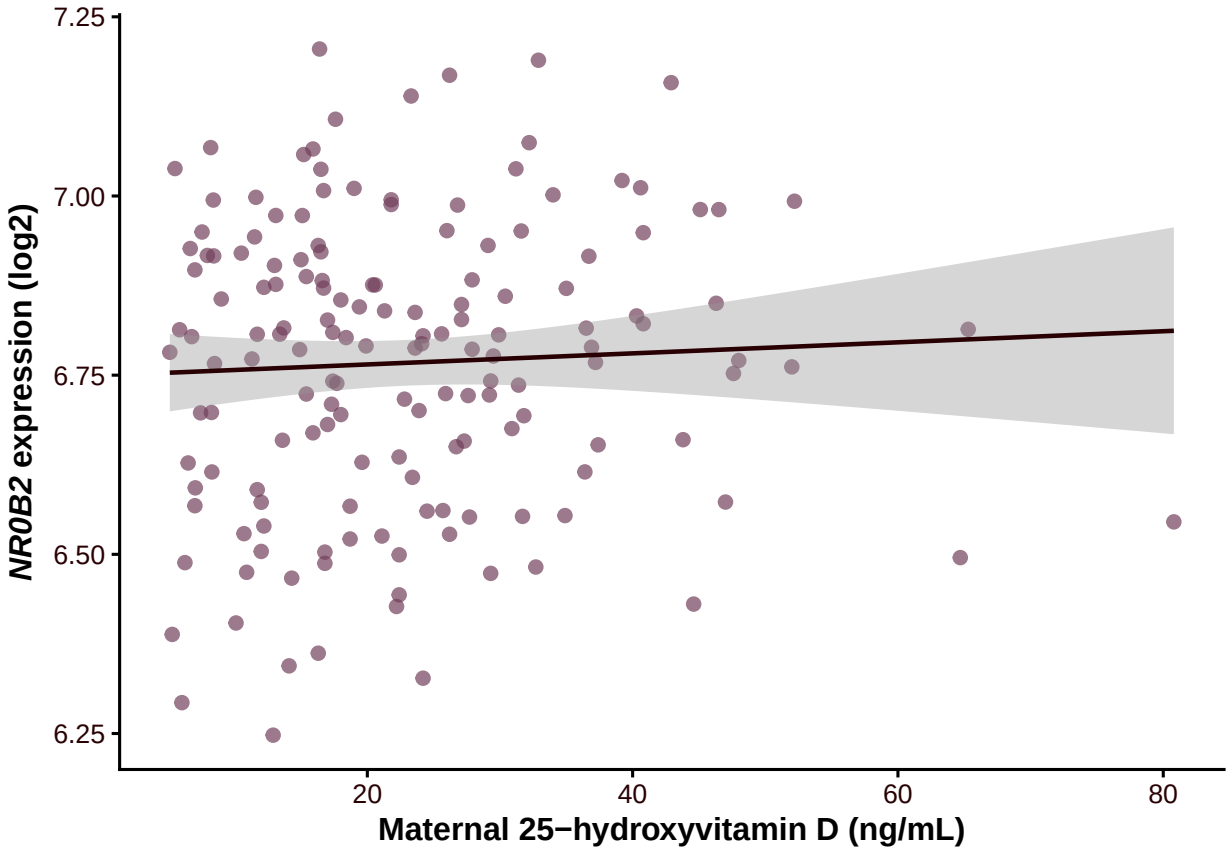


Figure 5: **Maternal vitamin D vs. *NR0B2* expression.** Each point is one participant (n = 157). The black line is a linear regression fit, and the gray band is its 95% confidence interval.

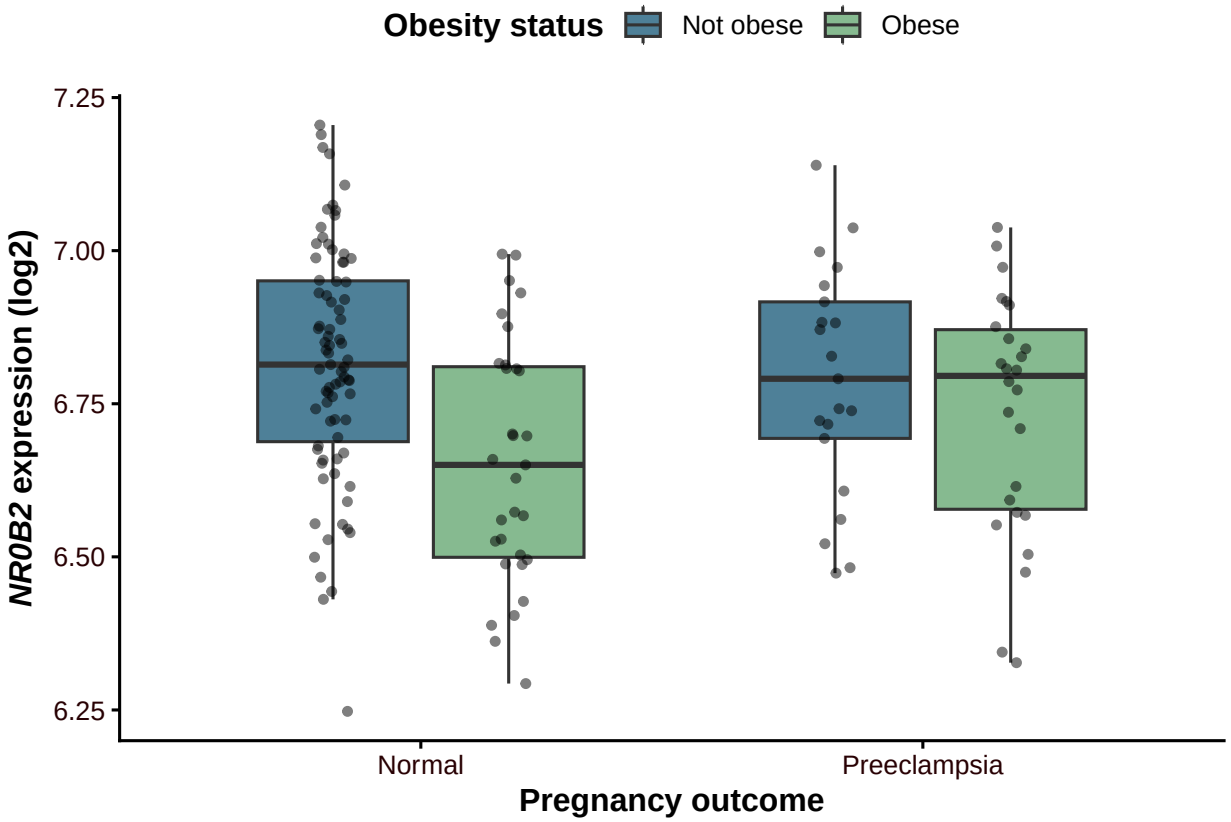


Figure 6: ***NR0B2* expression by pregnancy outcome and obesity status.** Obesity was defined as BMI $\geq$ 30 kg/m^2. Boxes show the median and interquartile range, whiskers extend to 1.5 $\times$ IQR, and points are individual participants. Group sizes: Normal/not obese n = 79, Normal/obese n = 31, Preeclampsia/not obese n = 21, Preeclampsia/obese n = 26.

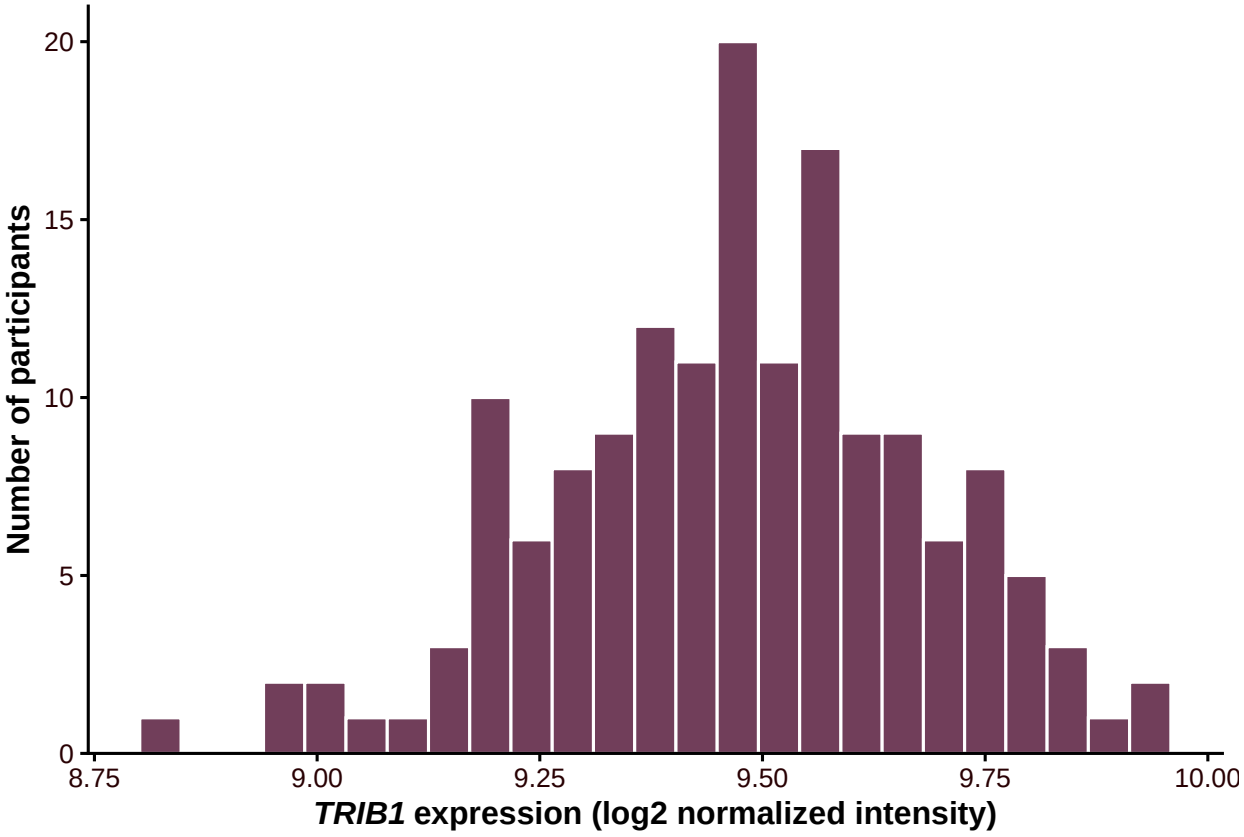


Figure 7: **Distribution of *TRIB1* expression.** Expression values from 157 pregnant participants are log2-transformed normalized intensities from the Affymetrix Human Gene 1.0 ST Array, measured in blood collected at 10–19 weeks' gestation.

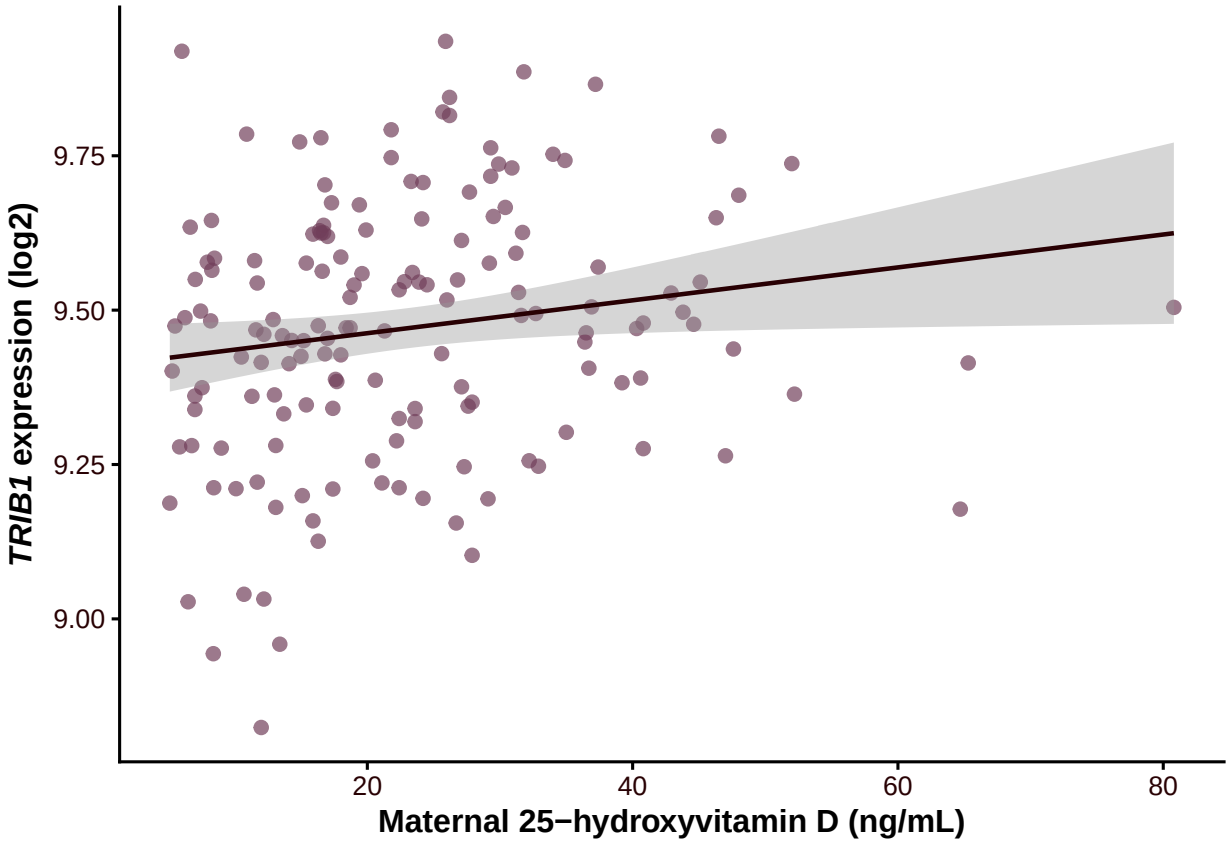


Figure 8: **Maternal vitamin D vs. *TRIB1* expression.** Each point is one participant (n = 157). The black line is a linear regression fit, and the gray band is its 95% confidence interval.

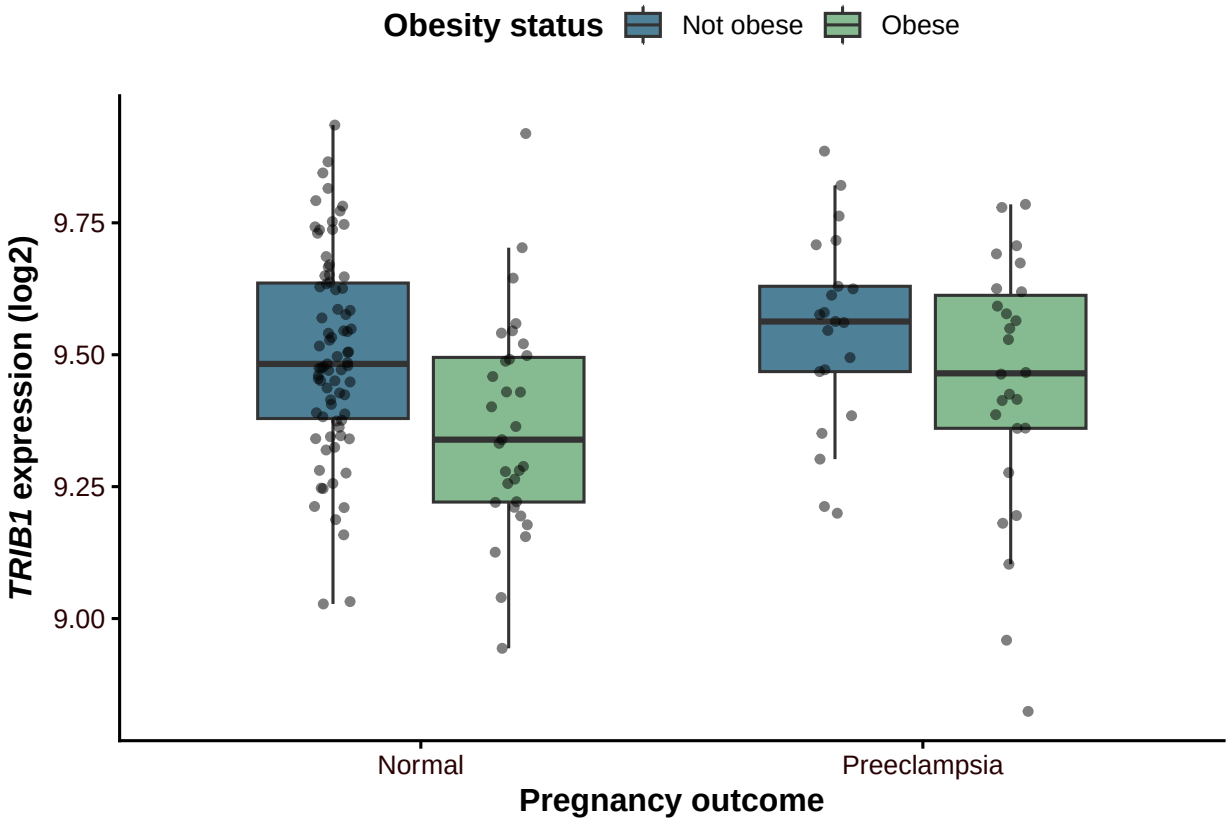


Figure 9: ***TRIB1* expression by pregnancy outcome and obesity status.** Obesity was defined as BMI $\geq$ 30 kg/m^2. Boxes show the median and interquartile range, whiskers extend to 1.5 $\times$ IQR, and points are individual participants. Group sizes: Normal/not obese n = 79, Normal/obese n = 31, Preeclampsia/not obese n = 21, Preeclampsia/obese n = 26.

```
            obese_pct = mean(obesity == "Obese") * 100)

# print rounded to one decimal place
summaryTable1 %>% mutate(across(where(is.double), ~ round(.x, 1)))
```

```
## # A tibble: 2 x 10
##   preg_preeclampsia     n age_mean age_sd gestAge_mean gestAge_sd vitD_mean
##   <fct>             <int>    <dbl>  <dbl>        <dbl>      <dbl>     <dbl>
## 1 Normal              110     26.6    5.1         39.3        1        24.4
## 2 Preeclampsia         47     25.8    4.9         37.8        3.5      19.7
## # i 3 more variables: vitD_sd <dbl>, obese_n <int>, obese_pct <dbl>
```

Heatmap

```
# Claude helped figure this code out

# 10 genes for the heatmap
heatGenes <- c("CXCR2", "LYN", "PRKCD", "AQP9", "FOSL2",
               "TRIB1", "NR0B2", "ACSL4", "EPHX2", "PEX12")

heatMat <- geneExp %>%
  filter(GENE %in% heatGenes) %>%      # keep only the 10 genes
  select(-PROBEID) %>%                 # remove the probe ID column
  column_to_rownames("GENE") %>%       # gene names become row names
  as.matrix()                          # convert to a numeric matrix

dim(heatMat) # expect 10 157
```

```
## [1]  10 157
```

```
# tracking-bar annotations: one row per participant, matched to heatMat columns
annotCol <- covar %>%
  select(geoID, preg_preeclampsia, obesity) %>%
  rename(Preeclampsia = preg_preeclampsia,   # nicer names for the legend
         Obesity = obesity) %>%
  column_to_rownames("geoID")

# put the rows in the same order as the heatmap columns
annotCol <- annotCol[colnames(heatMat), ]

# check the order matches exactly
all(rownames(annotCol) == colnames(heatMat)) # expect TRUE
```

```
## [1] TRUE
```

```
# colors for the tracking bars (reuse obesity colors so all figures match)
annotColors <- list(
  Preeclampsia = c("Normal" = "#D9D9D9", "Preeclampsia" = "#713E5A"),
  Obesity      = obesityColors
)

# color scale for expression: purple (low) -> white (average) -> orange (high)
heatColors <- colorRampPalette(c("#5E3C99", "white", "#E66101"))(100)

pheatmap(heatMat,
```

```
        scale = "row",                # z-score each gene across participants
        color = heatColors,
        clustering_distance_rows = "euclidean",
        clustering_distance_cols = "euclidean",
        clustering_method = "complete",
        annotation_col = annotCol,
        annotation_colors = annotColors,
        breaks = seq(-3, 3, length.out = 101),
        show_colnames = FALSE,        # hide participant IDs
        labels_row = as.expression(lapply(rownames(heatMat),
                                          function(x) bquote(italic(.(x))))),
        border_color = NA)
```

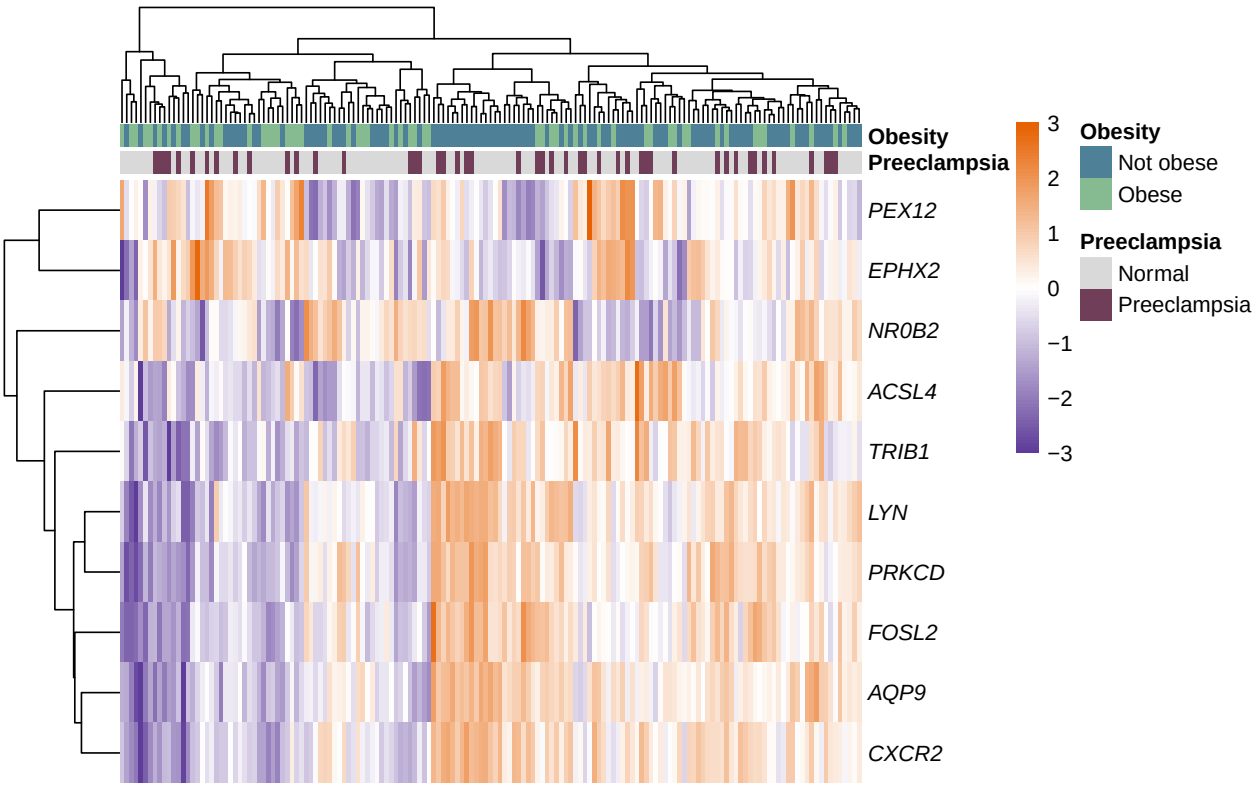


Figure 10: **Expression of ten inflammation- and metabolism-related genes across 157 participants.** Colors show gene expression z-scores (each gene scaled across participants; values beyond ±3 shown at the darkest color). Genes and participants were clustered by hierarchical clustering (Euclidean distance, complete linkage). Tracking bars show pregnancy outcome and obesity status (BMI $\geq$ 30 kg/m^2).

Forest plot

```
# Claude helped figure this code out
# vitamin D per 10 ng/mL so regression estimates are easier to read
datAll <- datAll %>% mutate(vitD10 = mat_vitD / 10)

# fit unadjusted and obesity-adjusted models for each gene,
# keeping the vitamin D estimate and 95% CI
forestDat <- data.frame()    # start with an empty data frame
```

```
for (g in heatGenes) {
  geneData <- datAll %>% filter(GENE == g)

  m1 <- lm(expression ~ vitD10, data = geneData)           # unadjusted
  m2 <- lm(expression ~ vitD10 + obesity, data = geneData) # adjusted for obesity

  forestDat <- rbind(forestDat,
    data.frame(gene = g, model = "Unadjusted",
               estimate = coef(m1)["vitD10"],
               lower = confint(m1)["vitD10", 1],
               upper = confint(m1)["vitD10", 2]),
    data.frame(gene = g, model = "Adjusted for obesity",
               estimate = coef(m2)["vitD10"],
               lower = confint(m2)["vitD10", 1],
               upper = confint(m2)["vitD10", 2]))
}

nrow(forestDat)   # expect 20
```

```
## [1] 20
```

```
# order genes top to bottom as in heatGenes, and fix the model order
forestDat <- forestDat %>%
  mutate(gene  = factor(gene, levels = rev(heatGenes)),
         model = factor(model, levels = c("Adjusted for obesity", "Unadjusted")))

forestPlot <- ggplot(forestDat, aes(x = gene, y = estimate, color = model)) +
  #line of no association
  geom_hline(yintercept = 0, linetype = "dashed", color = "#280003") +
  geom_pointrange(aes(ymin = lower, ymax = upper),
                  # dot= estimate, line = 95% CI
                  position = position_dodge(width = 0.4)) +
  coord_flip() +      # genes on the side
  scale_color_manual(values = c("Unadjusted" = "#713E5A",
                                "Adjusted for obesity" = "gray50"),
                     breaks = c("Unadjusted", "Adjusted for obesity"),
                     name = NULL) +
  scale_y_continuous(breaks = seq(-0.04, 0.12, by = 0.02)) +
  labs(x = NULL,
       y = "Change in expression (log2) per 10 ng/mL vitamin D (95% CI)") +

  theme_pub +
  theme(axis.text.y = element_text(face = "italic"))

forestPlot
```

```
ggsave("forest_plot.pdf", forestPlot, width = 7, height = 5)
```

Software Version

```
R.version.string
```

```
## [1] "R version 4.6.1 (2026-06-24)"
```

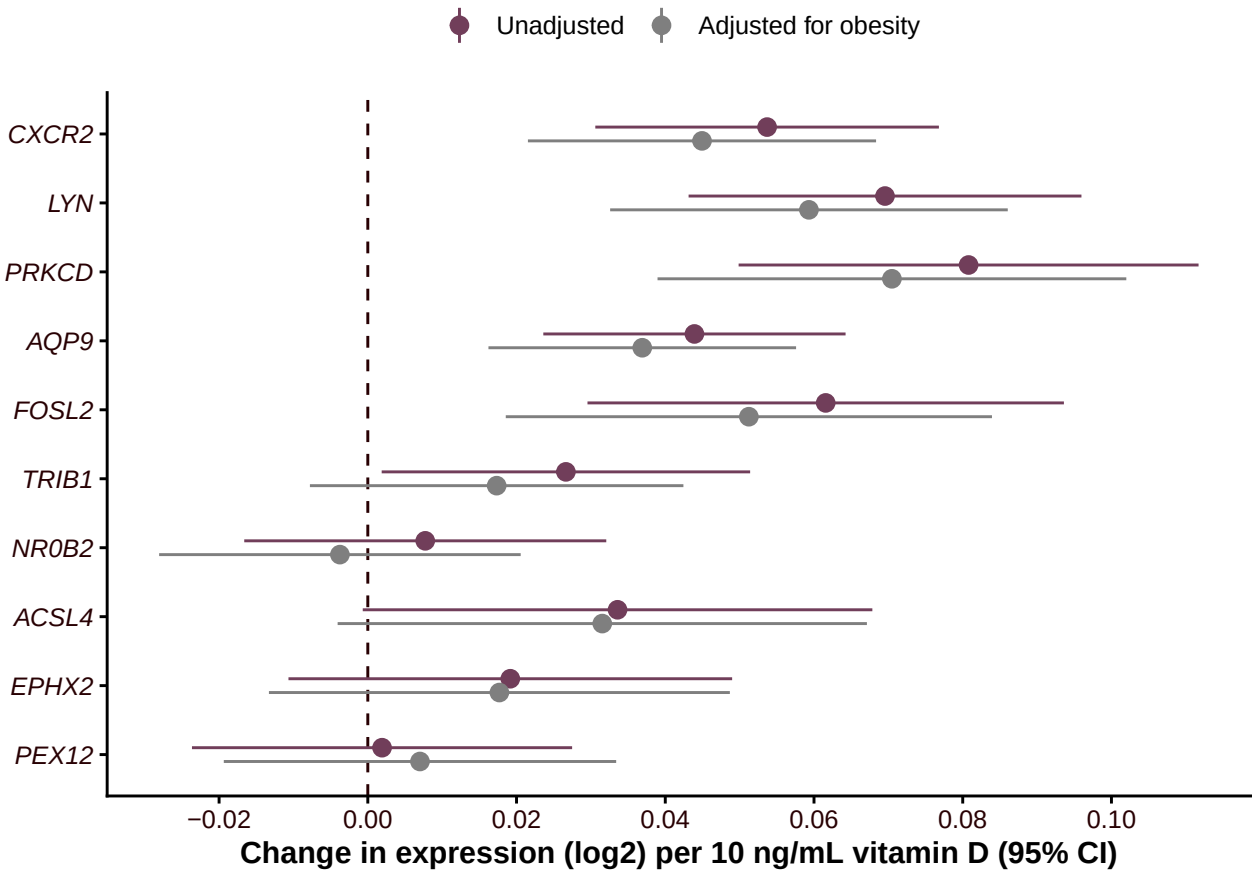


Figure 11: **Association between maternal vitamin D and expression of ten inflammation- and metabolism-related genes, before and after adjustment for obesity.** Points show the estimated change in log2 expression per 10 ng/mL increase in 25-hydroxyvitamin D from linear regression; lines show 95% confidence intervals. Purple: unadjusted model; gray: model adjusted for obesity (BMI $\geq$ 30 kg/m^2). The dashed line marks no association. n = 157.

```r
packageVersion("tidyverse"); packageVersion("ggplot2"); packageVersion("pheatmap")
```

```
## [1] '2.0.0'
## [1] '4.0.3'
## [1] '1.0.13'
```